Unit 8 Vitamins



Christiaan Eijkman

Christiaan Eijkman was Dutch physician who showed that the disease Beriberi was actually a nutritional defect. Interestingly, Eijkman discovery was accidental where he noted that chicken fed with polished rice developed Beriberi like symptoms, and feeding them back with unpolished rice made the chicken normal. This eventually led to the identification of the vitamin thiamine (B1). He was awarded Nobel prize along with Sir Fredrick Hopkins for the discovery of vitamins in 1929

Learning Objectives

After studying this unit the students will be able to:

- Distinguish fat and water soluble vitamins
- List the natural sources & dietary requirement of vitamins
- Describe the structure, function and the mode of storage of vitamins in our body
- Explain the different diseases associated with vitamin deficiencies
- Relate vitamins and co-enzyme functions.

Introduction

Vitamins are low molecular weight organic compounds indispensable for the normal activity of the organisms. Their absence causes deficiency symptoms. Most of the vitamins or their derivatives are coenzymes which are essential for many important enzymatic reactions in the cells.

Vitamins are required in small quantities for a variety of biochemical functions in our body. Most of them cannot be synthesised by the body and must be supplied through the diet. Unlike

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